

The Utility Problem

As electric distribution utilities aim to diversify and expand their energy generation mix, increased attention is being focused on the influx of alternate energy sources such as roof-mounted photovoltaic (PV) generation systems. This growing penetration of distributed PV generation in the grids poses new challenges for electricity distribution utilities, including their operations and planning. Distribution grid operators, especially, are now facing reduced and even reversed power flows at times of high PV generation and low load. This has a range of impacts, for example, voltage rise in the network. On the other hand, this situation brings some opportunities like the possibility of the provision of locally sourced carbon offsets or credits or the opportunity to identify the areas of the grids which might be operated in island operation during extreme cases.

For these reasons, knowledge of the PV generation per circuit (feeder) or substation is valuable information. Historically, a systematic approach to this insight is not yet widely implemented among distribution utilities. However, with the advancement of sensors and IoT devices in the grid, the data for this systematic approach are becoming available. Distribution utilities might take this opportunity and develop such a type of analysis better to understand the PV generation situation in the service area.

The Data Solutions and Methods

If a utility wants to conduct PV Generation analysis, multiple data sets are required. These datasets include the following:

1. GIS Data or other datasets defining energy consumers/producers/DERs on feeder or substation level.
2. Knowledge about the type of generation or DER.
3. Time series data from DERs like smart inverters, V2G chargers, batteries, etc.

With the Awesense Energy Transition Platform, conducting PV generation analysis is simplified since Awesense's Energy Data Model (EDM) interface provides numerous shortcuts. After the data ingestion to the platform, all required datasets are centralized, synchronized and validated, which is essential for easier development of analytics.

For building PV generation analysis on the Awesense Platform, the analyst should use the Awesense EDM and follow these simple steps:

1. Query the GIS dataset using the `get_connected()` geospatial tracing function to obtain:
 - a. Set of DERs (microgeneration or other DERs).
 - b. Set of energy consumers. (optional)All this can be readily available per feeder (circuit), eventually per substation.
2. By the definition of the EDM data model, the set of queried microgeneration and DER assets will contain information about the type of energy source. With this capability, choosing PV sources only is an easy task.

Registered PV Generation Analysis

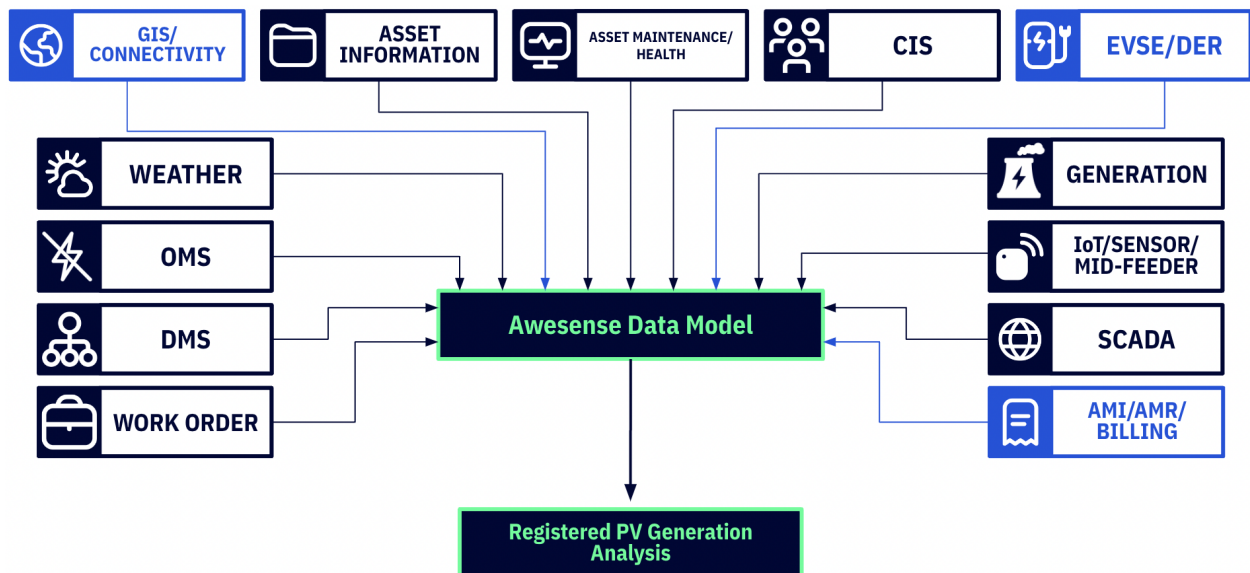
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3. Access the time-series data from:
 - a. DERs that are generating energy - a generation time series.
 - b. Energy consumer (electricity meters) consumption - a consumption time series.
4. Sum the total generation per feeder (eventually per substation) and sum the total consumption per feeder (eventually per substation).
5. Calculate the total generation per circuit (or optionally per substation) or quantify the % of the circuit's load covered by microgen/DER generation.

Required Data Sets

The required data for this use case are visualized in the following diagram. Data sources highlighted in [royal blue](#) are required for this use case.



Available Use Case Implementations:

Users of the Awesense Energy Transition Platform can access this use case by the following means:

1. GitHub [link](#) to the Jupyter notebook can be provided upon request.